

Class Activity- I/O Operations -CO5

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COURSE CODE: CSA0904

COURSE NAME: PROGRAMMING FOR JAVA

1 Create a program to create a file and write the data

```
import java.io.FileWriter;
```

```
import java.io.IOException;
```

```
import java.util.Scanner;
```

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        try {
```

```
            FileWriter file = new FileWriter("data.txt");
```

```
            System.out.print("Enter data to write into the file: ");
```

```
            String data = sc.nextLine();
```

```
            file.write(data);
```

```
            file.close();
```

```
            System.out.println("File created successfully!");
```

```
            System.out.println("Data written into data.txt");
```

```

    } catch (IOException e) {

        System.out.println("An error occurred.");

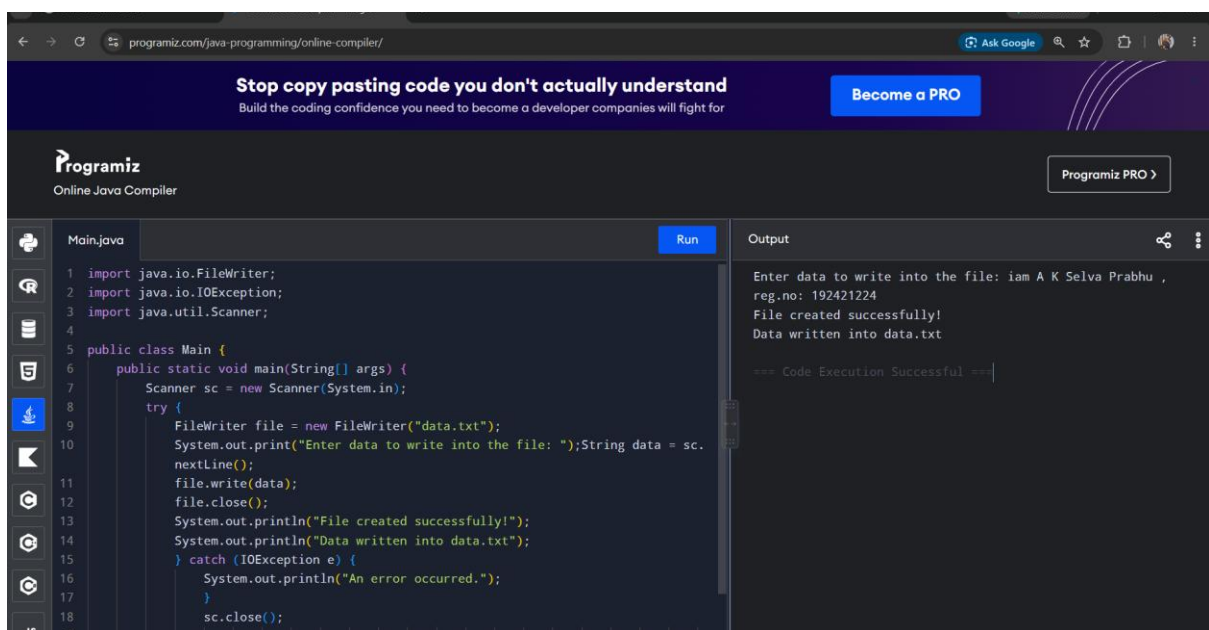
    }

    sc.close();

}

}

```



2 Read and Write using FileReader/FileWriter Class

```

import java.io.FileReader;

import java.io.FileWriter;

import java.io.IOException;

```

```

public class Main {

    public static void main(String[] args) {

        try {

```

```
// Write data into file

FileWriter writer = new FileWriter("data.txt");

writer.write("Hello Java Programming");

writer.write("\nThis is FileReader and FileWriter example.");

writer.close();


// Read data from file

FileReader reader = new FileReader("data.txt");


int ch;

while ((ch = reader.read()) != -1) {

    System.out.print((char) ch);

}


reader.close();

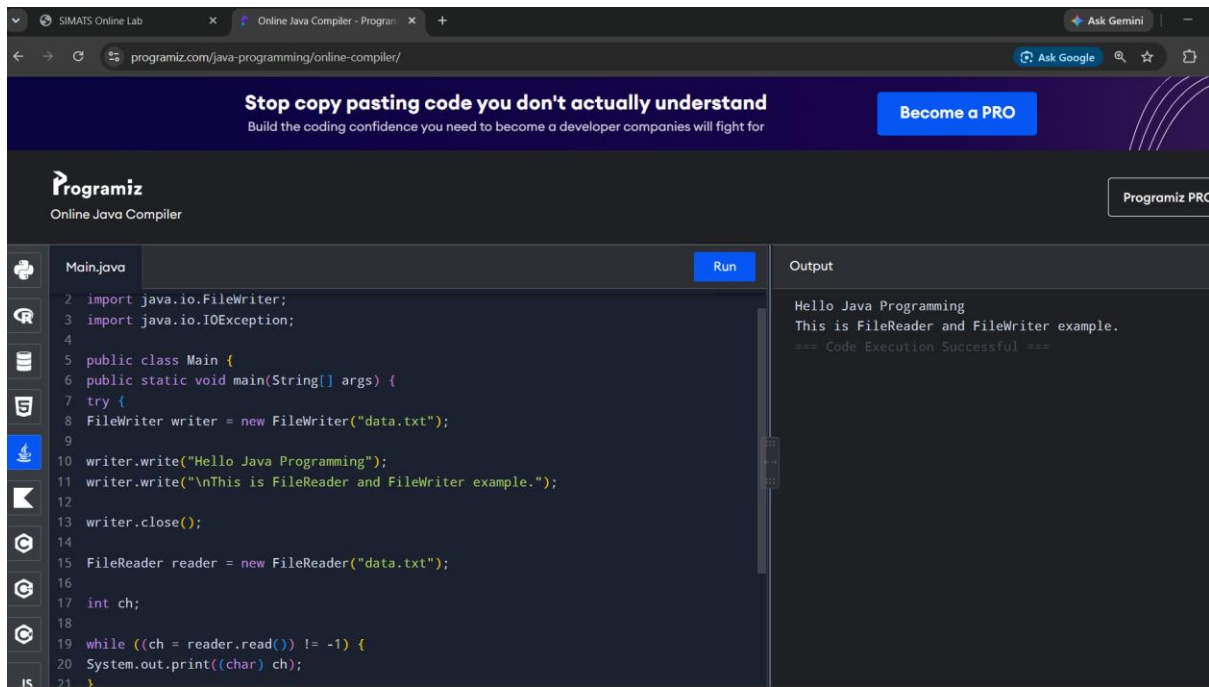

} catch (IOException e) {

    System.out.println("Error: " + e.getMessage());

}

}

}
```



3 Copy the content from source to destination using BufferedReader and BufferedWriter

```
import java.io.BufferedReader;
```

```
import java.io.BufferedWriter;
```

```
import java.io.FileReader;
```

```
import java.io.FileWriter;
```

```
import java.io.IOException;
```

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        try {
```

```
            FileWriter fw = new FileWriter("source.txt");
```

```
            fw.write("Hello Java Programming\n");
```

```
            fw.write("This content is copied to another file.");
```

```
            fw.close();
```

```
            BufferedReader reader = new BufferedReader(new FileReader("source.txt"));
```

```
BufferedWriter writer = new BufferedWriter(new FileWriter("destination.txt"));
```

```
String line;
```

```
while ((line = reader.readLine()) != null) {
```

```
    writer.write(line);
```

```
    writer.newLine();
```

```
}
```

```
reader.close();
```

```
writer.close();
```

```
System.out.println("File copied successfully!");
```

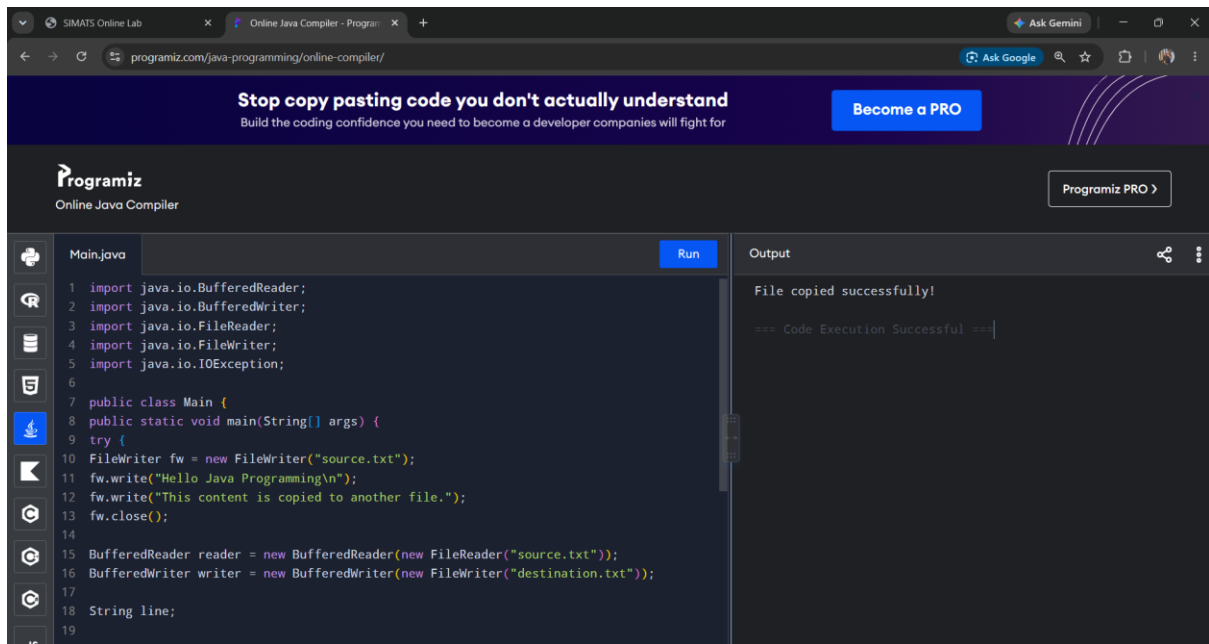
```
} catch (IOException e) {
```

```
    System.out.println("Error: " + e.getMessage());
```

```
}
```

```
}
```

```
}
```



4 Count number of characters and lines in the given file, use this simple code

```
import java.io.FileWriter;
```

```
import java.io.BufferedReader;
```

```
import java.io.FileReader;
```

```
import java.io.IOException;
```

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        try {
```

```
            FileWriter writer = new FileWriter("source.txt");
```

```
            writer.write("Hello Java\n");
```

```
            writer.write("File Handling\n");
```

```
            writer.write("Programming");
```

```
            writer.close();
```

```
BufferedReader reader = new BufferedReader(new FileReader("source.txt"));
```

```
int characters = 0;
```

```
int lines = 0;
```

```
String line;
```

```
while ((line = reader.readLine()) != null) {
```

```
    lines++;
```

```
    characters += line.length();
```

```
}
```

```
reader.close();
```

```
System.out.println("Number of lines: " + lines);
```

```
System.out.println("Number of characters: " + characters);
```

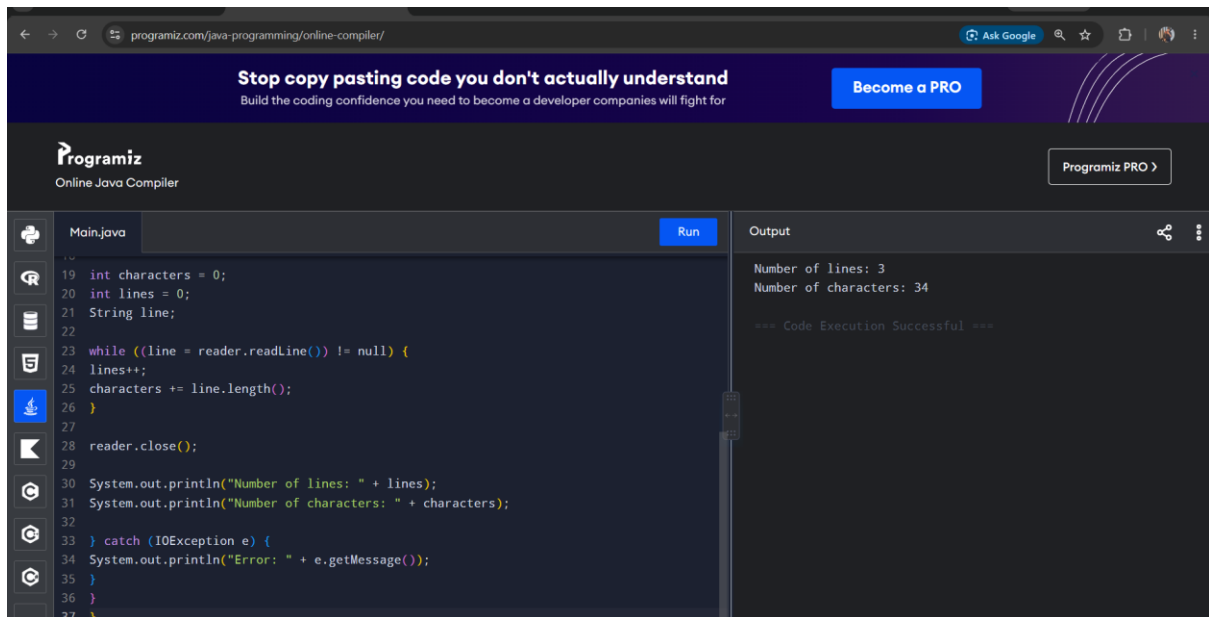
```
} catch (IOException e) {
```

```
    System.out.println("Error: " + e.getMessage());
```

```
}
```

```
}
```

```
}
```



5 Delete a file, if it exists, and create it again, use this simple code.

```
import java.io.File;
```

```
import java.io.IOException;
```

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        try{
```

```
            File file = new File("sample.txt");
```

```
            if (file.exists()) {
```

```
                file.delete();
```

```
                System.out.println("Old file deleted.");
```

```
            }
```

```
            if (file.createNewFile()) {
```

```
                System.out.println("New file created.");
```

```
            }
```



```

} catch (IOException e) {

System.out.println("Error: " + e.getMessage());

}

}

}

```

